

Solution

WORK-04

Class 09 - Science

Section A

1. State True or False:

(i) (a) True

Explanation: True

(ii) (b) False

Explanation:

False. The portion of the medium where density (or pressure) has a value larger than its average value is called a crest.

(iii) (a) True

Explanation: True

(iv) (a) True

Explanation: True

(v) (b) False

Explanation: False. Sound waves cannot travel through a vacuum but light waves and radio waves can travel through a vacuum.

(vi) (a) True

Explanation: True

(vii) (a) True

Explanation: True

(viii) (b) False

Explanation: False. The change in density (or pressure) from the maximum value to the minimum value and again to the maximum value is called an oscillation.

(ix) (a) True

Explanation: True

(x) (a) True

Explanation: True

(xi) (a) True

Explanation: True

(xii) (a) True

Explanation: True

(xiii) (a) True

Explanation: True

(xiv) (b) False

Explanation: False. Compression is the region of high pressure and rarefaction is the region of low pressure.

(xv) (a) True

Explanation: True

(xvi) (a) True

Explanation: True

(xvii) (a) True

Explanation: True

(xviii) (b) False

Explanation: False. A sound of single frequency is called a tone.

(xix) **(a)** True

Explanation: True

(xx) **(b)** False

Explanation: False. The number of complete oscillations per second is called the frequency (ν) of the sound wave.

Section B

2. Fill in the blanks:

(i) 1. Wavelength

(ii) 1. Amplitude

(iii) 1. Quality

(iv) 1. Reverberation

(v) 1. Sound

(vi) 1. Pitch

(vii) 1. Note

(viii) 1. Echo

(ix) 1. Medium

(x) 1. Increases

(xi) 1. Vibration

(xii) 1. Ultrasound

(xiii) 1. Wave

(xiv) 1. Air

(xv) 1. Oscillation

(xvi) 1. Physiological

(xvii) 1. Loudness

Section C

3.

(d) 3000 m

Explanation: Distance travelled by wave = $2d$

Time taken = 4sec

speed = 1500m/s

so, $2d = 1500 \times 4$

$d = 3000\text{m}$

4.

(d) A

Explanation: Angles are always measured with the normal at the point of incidence of reflection.

5. **(a)** Detector

Explanation: The detector converts the ultrasonic waves into electrical signals which are appropriately interpreted.

6.

(c) Pitch

Explanation: Pitch is the physiological sensation that helps in distinguishing a shrill sound from a flat sound. The faster the vibration of the source, the higher is the frequency and the higher is the pitch.

7. **(a)** C

Explanation: Seems more proper, since a perpendicular line can give $\angle r = \angle i$.

8.

(c) (ii)

Explanation: Reflection brings reaction to the action of a wave.

9.

(b) (i)

Explanation: Angle with normal is always measured.

10. (c) Reflection of sound
Explanation: The image shows the reflection of sound. The directions in which the sound is incident and is reflected make equal angles with the normal to the reflecting surface, and the three are in the same plane.
11. (a) Longitudinal waves
Explanation: The wave formed because of the oscillation; parallel to the disturbance; is called longitudinal wave. Longitudinal wave are formed compressions and rarefactions.
12. (a) Multiple reflection
Explanation: The sound of heartbeats reaches the doctor's ears by the multiple reflections of the sound waves through the stethoscope tube.
13. (a) backward and forward along the length of the slinky
Explanation: Longitudinal wave is formed.
14. (c) Multiple reflection
Explanation: The soundboard works on the principle of multiple reflections of sound. The concave surface of the soundboard reflects the sound waves of the speaker towards the audience due to which sound is distributed uniformly throughout the hall and even the persons sitting at the back of the hall can hear his speech easily.
15. (a) All of these
Explanation: Sonar is a device that uses ultrasonic waves to measure the distance, direction and speed of underwater objects.
16. (b) 4 m/s
Explanation: $\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{20}{5} = 4 \text{ m/s}$
17. (a) D
Explanation: $\angle r = \angle i$ at the point of incidence
18. (b) B D
Explanation: As one wavelength = one crest + one trough,
 Half the wavelength = one crest or one trough = AC (or CE).
 Further, AC = AB + BC = BC + CD = BD (as AB = CD)
19. (a) $\frac{\lambda}{2}$
Explanation: Wavelength is the length between two consecutive peaks, i.e. crest or two consecutive valleys, i.e. trough of a wave. Wavelength is represented by Greek letter λ (lambda). So, the separation between T and C is $\frac{\lambda}{2}$.
20. (a) $P_A > P_B$
Explanation: The sensation of a frequency is commonly referred to as the pitch of a sound. A high pitch sound corresponds to a high frequency sound wave and a low pitch sound corresponds to a low frequency sound wave.
21. (b) A
Explanation: Angle with normal is to be measured.
22. (a) Echolocation
Explanation: Bats search their prey at night by the method of echolocation. They emit high-frequency ultrasonic squeaks while flying and listen to the echoes produced by the reflection of their squeaks from the prey like a flying object. From the time taken by the echo to be heard, bats can judge the distance of the insect and hence catch it.
23. (b) (ii)
Explanation: Reaction is always equal and opposite to action.
24. (b) B, X
Explanation: Cross-section not to diffuse the sound and a good reflecting surface are the need.

Section D

25. **(b) Statement A**
Explanation: Bat and porpoises use ultrasound for navigation and location of food in the dark. They contain ultrasound producing organs and receiver that help them to judge the position and location of food and predator. Music and TV sounds that we hear have a noise level below 90 decibels.
26. **(c) Statement B**
Explanation: The speed that is greater than the speed of sound is known as supersonic. The amount of sound energy, passing each second through unit area is called intensity of sound. So, statement B is a true statement and statement A is a false statement.
27. **(b) Both A and B**
Explanation:
 - Human beings are ultrasound deaf as humans are not able to hear the frequency of sound above 20,000Hz.
 - Ultrasound travels through various media including gases, liquids, and solids, but cannot travel through a vacuum. The speed of sound varies by the medium it travels through. Sound is likely to travel faster through solids, followed by liquids and gases.
28. **(a) Statement B**
Explanation: Wave motion is an activity that carries energy from one place to another without actually moving any matter. In wave motion, energy-the ability to perform work, or to exert force over distance-is transmitted from one place to another without actually moving any matter along the wave. In some types of waves, such as those on the ocean, it might seem as though matter itself has been displaced; that is, it appears that the water has actually moved from its original position. In fact, this is not the case: molecules of water in an ocean wave move up and down, but they do not actually travel with the wave itself. Only the energy is moved. Electromagnetic waves differ from mechanical waves in that they do not require a medium to propagate. This means that electromagnetic waves can travel not only through the air and solid materials but also through the vacuum of space.
29. **(a) Statement B**
Explanation: An object moving at a speed greater than that of sound is moving with supersonic speed and not ultrasonic speed. Wave motion transfers energy from one point to another, which displace particles of the transmission medium—that is, with little or no associated mass transport. Waves consist, instead, of oscillations or vibrations (of a physical quantity), around almost fixed locations.
30. **(d) Both A and B**
Explanation: The speed of sound increases with temperature. So, sound travels faster in summer than in the winter season. At a constant temperature, the gas pressure has no effect on the speed of sound, since the density will increase, and since pressure and density (also proportional to pressure) have equal but opposite effects on the speed of sound, and the two contributions cancel out exactly.
31. **(a) Both A and B**
Explanation: Speed of sound is different at different temperature. With increase in temperature, speed of sound increases. The loudness of sound is measured in units called decibels (dB). A decibel unit expresses the relative intensity of sounds on a scale from zero for the average least perceptible sound to about 100 dB, which is near the level most people find uncomfortably loud.

Section E

32. **(d) A**
Explanation:
 - Sound waves are longitudinal waves that require material medium for its propagation.
 - Moon does not have an atmosphere for the movement of sound waves.
 - The speed of sound is different at different temperatures.

- Ultrasound shock waves are used to break up a kidney stone.

33. (a) B and C

Explanation: The SI unit of wavelength is metre (m) and that of velocity is ms^{-1} .

34.

(b) (A)

Explanation: The speed of a sound wave depends on the properties of the medium through which it moves and the only way to change the speed is to change the properties of the medium. So, given statement is incorrect statement.

35. (a) Both statement (B) and (C)

Explanation: Rolling thunder is repeated claps of thunder in quick succession. The sound of the same loudness and pitch produced by two different sources can be distinguished by timbre. Female's voice has more frequency components compared to men. Women speak at one octave higher than men.

So, statement (A) is incorrect and (B) and (C) are correct statements.

36.

(c) B and C are incorrect

Explanation: SI unit of wavelength is the metre (m) and velocity is ms^{-1} .

37.

(b) (A)

Explanation: Sound waves are mechanical and elasticity based and hence longitudinal. Light waves are electromagnetic and are transverse.